

The ANNNI phase diagram under CNOT-target depolarizing noise

Benjamin Kaganovitch, Dimitrii Kashtanov, Yan Ivanov, Frederick Pu

Q-SITE Hacks 2026 • Quantum Coalition Scientific Open Challenge

Code, data and notebook: github.com/BenjaminKaganovitch/qsites-2026-open-challenge

Abstract. We map the ground-state phase diagram of the periodic ANNNI chain for $\kappa \in [0, 1]$, $h \in [0, 2]$ with sparse exact diagonalisation at $N = 8, 10, 12$ and with a symmetry-preserving Hamiltonian-variational circuit (128 CNOTs) that reproduces the exact phase labels at 98.9% of grid points. We then evaluate the same circuit with a depolarizing channel on the target after every CNOT. At $p = 0.01$ the order parameters contract to about 60% of their clean values and the ferromagnet disappears under fixed thresholds, while boundaries defined by derivative peaks move by a median of only 0.005; at $p = 0.05$ the state is essentially maximally mixed. Channel-insertion sums explain why the ferromagnet is the most fragile phase for this circuit, and zero-noise extrapolation recovers 98% of the $p = 0.01$ labels but fails at $p = 0.05$. Every number below comes from the saved analysis outputs in the repository. AI coding assistants were used, as permitted by the organisers.

1 Problem and conventions

1.1 Model and diagnostics. We study the axial next-nearest-neighbour Ising (ANNNI) chain

$$H = - \sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i, \quad (1)$$

with Pauli operators, $J_1 = 1$ and periodic boundary conditions, over $\kappa \in [0, 1]$ and $h \in [0, 2]$. Because the ground state of a finite symmetric ring keeps the global spin-flip symmetry, the one-point magnetisation $\langle Z_i \rangle$ vanishes even in the ordered phases [S07], so we identify the phases from longitudinal correlations instead. We use the structure factor

$$S(q) = \frac{1}{N} \sum_{i,j} e^{iq(i-j)} \langle Z_i Z_j \rangle, \quad (2)$$

with the self-terms kept, and define the floor-subtracted order parameters $O_F = (S(0) - 1)/(N - 1)$ and $O_A = (S(\pi/2) - 1)/(N/2 - 1)$, which equal 1 for perfect order and 0 for an uncorrelated state. The transverse magnetisation m_x completes the set. We checked these definitions on ideal states: the ferromagnet gives $S(0) = N$, the antiphase gives $S(\pi/2) = N/2$, and both $|+\rangle^{\otimes N}$ and $I/2^N$ give $S(q) = 1$, with m_x telling the last two apart.

1.2 Reference curves. The curves overlaid on the diagrams are approximate thermodynamic references rather than finite-ring labels [S10, S13]. They are the second-order Ising estimate (rationalised so that $h_I(0) = 1$), the lower antiphase-floating fit $h_{PT} = 1.05(\kappa - \frac{1}{2})$, which describes a Pokrovsky–Talapov transition [S12], and the upper Kosterlitz–Thouless (KT) fit $h_{KT} = 1.05\sqrt{(\kappa - \frac{1}{2})(\kappa - 0.1)}$.

2 Methods

2.1 Exact reference. SciPy’s sparse Lanczos solver returns the lowest six eigenpairs on 31×31 grids for $N = 8, 10, 12$, and on fine cuts of 21 values of κ by 401 values of h for all three sizes. Warm-started sweeps in h agree with independent

cold starts to 10^{-13} . Degenerate ground spaces are treated as equal mixtures, and fidelities are measured between ground subspaces. The Hamiltonian matches the starter-kit exact diagonalisation and PennyLane’s matrix representation to 7×10^{-15} .

2.2 Classifier. The thresholds are set from anchor regions fixed by limiting-case physics rather than from the overlay curves: the ferromagnet at $\kappa \leq 0.2, h \leq 0.2$; the antiphase at $\kappa \geq 0.8, h \leq 0.1$; and the paramagnet at $h \geq 1.8$. This gives $T = 0.541$ for the order parameters and $T_x = 0.490$ for m_x . A point is labelled ferro or antiphase if $\max(O_F, O_A) \geq T$, paramagnet if that fails but $m_x \geq T_x$, and unpolarised otherwise. Points whose label flips when the thresholds are moved by ± 0.1 are marked ambiguous. We also report a threshold-free relative-dominance label, $\arg \max(O_F, O_A, m_x^2)$. Under uniform terminal depolarisation all three quantities contract by the same factor a^{2m} with $a = 1 - 4p/3$ [S28], so in that benchmark this label is invariant under noise, and any change it does show reveals circuit-specific noise action.

2.3 Circuit. We use a Hamiltonian-variational ansatz (HVA) [S30]: the product state $|+\rangle^{\otimes N}$ followed by $L = N/2 = 4$ layers of

$$U_\ell = e^{-ib_\ell \sum_i X_i} e^{-ic_\ell \sum_i Z_i Z_{i+2}} e^{-ia_\ell \sum_i Z_i Z_{i+1}}, \quad (3)$$

which gives 12 parameters and preserves translation and parity symmetry. Each ZZ rotation compiles to CNOT–RZ–CNOT, so the circuit contains 128 CNOTs in total. A hardware-efficient RY/CNOT ring was tried first and stalled at 1–3% energy error near the phase boundaries even at depth 6. The parameters are optimised at $p = 0$ by L-BFGS with exact adjoint gradients, using forward and reverse warm-started sweeps and random restarts. Noise is PennyLane’s `DepolarizingChannel(p)` applied to the target qubit after every CNOT on the `default.mixed` simulator; the 128 channels were verified on the circuit tape. A NumPy density-matrix engine that matches `default.mixed` to 3×10^{-14} runs the zero-noise-extrapolation and response sweeps.

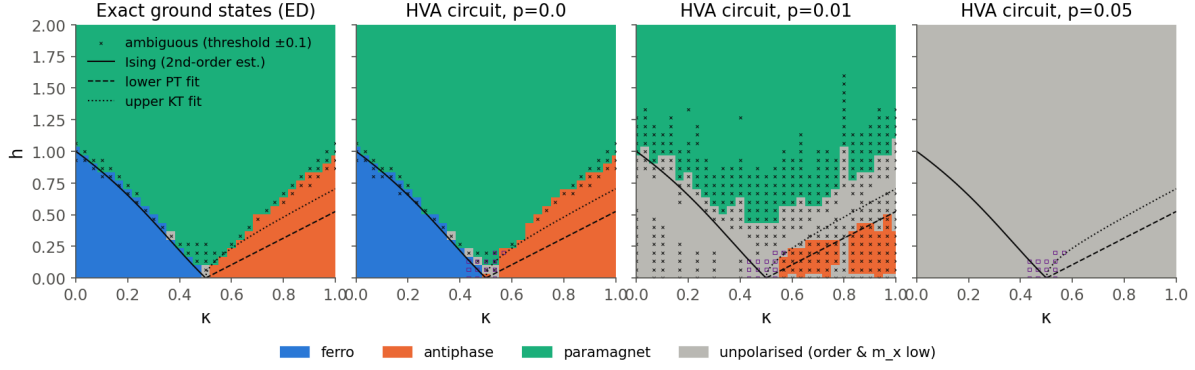


Figure 1. Phase diagrams on the 31×31 grid ($N = 8$, periodic). From left to right: exact ground states, then the HVA circuit at $p = 0, 0.01$ and 0.05 with fixed parameters and fixed thresholds. Purple squares mark points where the clean preparation fidelity is below 0.9, and crosses mark threshold-ambiguous points. The curves are the Ising estimate (solid), the lower PT fit (dashed) and the upper KT fit (dotted).

2.4 Protocol. The main experiment uses fixed parameters: the same parameters, circuit, grid, thresholds and colour scales are used at $p = 0, 0.01$ and 0.05 . The noise-induced shift $\delta h(\kappa, p) = h_b(p) - h_b(0)$ is measured for the same circuit, and the preparation bias $h_b(0) - h_b^{\text{ED}}$ is reported separately.

3 Clean results and finite size

3.1 Circuit accuracy. At $p = 0$ the circuit is an accurate stand-in for the exact ground states. The median relative energy error is 4×10^{-5} and the median infidelity $1 - F$ is 5×10^{-5} , where F is the ground-subspace fidelity. Only 1.5% of grid points have $F < 0.9$, all of them around the multicritical point $\kappa \approx 0.5$, and the circuit labels agree with exact diagonalisation at 98.9% of points (Figure 1). The preparation bias of the threshold boundary is at most 0.034 outside $0.42 < \kappa < 0.6$ and reaches 0.066 at $\kappa = 0.57$, next to the multicritical point. Two independent optimiser seeds agree on this boundary to a median of 0.007 and a maximum of 0.05.

3.2 Finite-size behaviour. The boundary between the ferromagnet and the paramagnet is almost independent of system size and follows the Ising estimate. At $\kappa = 0.2$ the fidelity-susceptibility peak moves from 0.616 at $N = 8$ to 0.629 at $N = 12$, towards the DMRG value of 0.639 reported in [S10]; the second-order estimate is 0.653. The antiphase boundary, by contrast, drifts strongly with size. At $\kappa = 0.75$ the order crossing falls from 0.562 at $N = 8$ to 0.432 at $N = 12$, approaching the KT fit at 0.423 from above. The ring with $N = 10$ cannot hold a period-4 pattern and is visibly anomalous: at $\kappa = 0.6$ its crossing sits at 0.397, against 0.244 for $N = 8$ and 0.164 for $N = 12$. This is a commensurability effect of the finite ring, not physics of the thermodynamic

limit.

3.3 Floating phase. At $N = 12$ the half-chain entropy of the pure ground state peaks close to the KT fit at $\kappa = 0.75$ and $\kappa = 0.9$, and the antiphase boundary drifts towards the KT line as N grows. Both observations are compatible with a critical region. However, the dominant wavevector q^* stays locked at $\pi/2$ between the PT and KT fits at every size we can reach. The only jumps in q^* occur above the KT line, inside the modulated paramagnet [S10], and the momentum resolution of $\Delta q = \pi/6$ at $N = 12$ cannot resolve an incommensurate modulation. We therefore do not claim a floating phase [S12, S23]. As a consistency check rather than an independent validation, a label-free k -means clustering on $(S(q)/N, m_x)$ reproduces the clean rule labels at 99.1% of points.

4 Noise: effects, magnitude and mechanism

4.1 Noise level $p = 0.01$. The order parameters contract to a median of 0.60 of their clean values, compared with 0.65 if every target hit acted as end-of-circuit noise. The contraction depends on the phase. In the limit $h \rightarrow 0$, O_F at $\kappa = 0.2$ drops from 1.00 to 0.37, whereas O_A at $\kappa = 0.9$ drops only to 0.59. Under the fixed threshold the ferromagnet disappears entirely: none of the points that exact diagonalisation labels ferro keep their label, while 46% of antiphase points and 89% of paramagnet points survive. Wherever a threshold crossing does survive it moves to lower h , by 0.04 to 0.48. This has the same sign as the exact first-order prediction $-p \partial_p O / \partial_h O$ but is 1.5 to 4 times larger, because the response is already nonlinear at this noise level. The boundaries defined by derivative peaks, on the other hand, barely move (Table 1): the median $|\delta h|$ is 0.005, and 28 of the 31 rows stay within 0.03, which is half a grid step. The exceptions are $\kappa = 0.13$

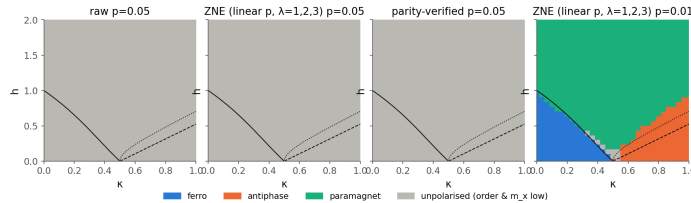


Figure 2. Error mitigation with fixed thresholds. From left to right: the raw map at $p = 0.05$, zero-noise extrapolation at $p = 0.05$, parity verification at $p = 0.05$, and zero-noise extrapolation at $p = 0.01$ (Richardson, $\lambda = 1, 2, 3$). Extrapolation restores the $p = 0.01$ diagram almost fully, while at $p = 0.05$ no usable signal remains.

Table 1. Boundary positions and noise-induced shifts for the HVA circuit. h_T is the fixed-threshold crossing and h_D the position of the maximum of $-dO/dh$. The grid step is 0.067; the seed-to-seed spread of δh_D at $p = 0.01$ has median 0.011 and maximum 0.038. The full table for all 31 values of κ is in the repository.

κ	Order	h_T^{ED}	Prep. bias	$h_D(0)$	$\delta h_D(p=0.01)$	$\delta h_D(p=0.05)$	h_T at $p=0.01$
0.2	ferro	0.685	-0.000	0.653	+0.027	+0.64 [†]	absent
0.3	ferro	0.487	-0.016	0.469	+0.015	+0.19	absent
0.4	ferro	0.246	+0.000	0.248	+0.003	+0.006	absent
0.7	anti	0.467	+0.018	0.477	-0.001	-0.004	absent
0.8	anti	0.650	+0.024	0.672	+0.003	+0.045	absent
0.9	anti	0.809	+0.004	0.723	+0.000	+0.098	absent

[†] Unreliable: the order signal at $p = 0.05$ is too small to locate a derivative peak.

and 0.27, which shift by +0.04, and $\kappa = 0.93$, which shifts by -0.12. This is the behaviour that [S28] predicts for a near-uniform rescaling of the observables, and accordingly the relative-dominance map at $p = 0.01$ agrees with the clean exact-diagonalisation map at 97% of points, compared with 99% at $p = 0$.

4.2 Noise level $p = 0.05$. At this noise level the state is essentially maximally mixed: the median purity is 0.0047, against 0.0039 for $I/2^8$. Every point is unpolarised under the fixed thresholds. In the relative-dominance map it is the paramagnet that shrinks, because m_x keeps only 16% of its clean value and hence m_x^2 keeps about 2.6%, while the order parameters keep a median of 8.5%.

4.3 Mechanism from exact channel-insertion sums. We evaluated the first-order response $\partial_p \langle O \rangle = \sum_g \text{Tr}\{O_g \mathcal{L}_t(\rho_g)\}$ one channel location at a time [S28]. For the ferromagnetic state $d \ln O_F / dp = -99$, for the antiphase $d \ln O_A / dp = -69$, and for m_x in the paramagnet the value lies between -32 and -48. The value that end-of-circuit noise would give for pair correlators is -43. Errors that occur mid-circuit are carried through the later CNOT-RZ-CNOT gadgets and damage the ZZ order more than errors at the end of the circuit would. The early layers contribute the most: for the ferro point, dO_F / dp receives -27 from layer 1 and -17 from layer 4. One consistent reading of why the ferromagnet is the most fragile phase is that its GHZ-like finite-ring ground state loses all $N(N-1)$ positive correlators in O_F to a single propagated X or Y error, whereas O_A mixes signs and the damage partly cancels. This ranking is specific to this circuit and to these diagnostics, and it reverses the organiser’s hypothesis [S01]. The noise does not change the spectrum of the Hamiltonian; what these maps show is the degradation of the prepared state.

4.4 Depth trade-off. We repeated the fixed-parameter experiment with $L = 1$ to 4 layers on four cuts in κ . The clean error $\langle |O - O^{\text{ED}}| \rangle_h$ falls with depth while the noise error grows with it, so the best depth shrinks as p grows. The antiphase is already accurate at $L = 2$, with a clean error of 0.024 at $\kappa = 0.8$, and $L = 2$ remains the best depth there at both $p = 0.01$ and $p = 0.05$. The ferromagnet needs $L = 4$ to be accurate, since its error at $L = 2$ is 0.17, and at $p = 0.01$ its best depth is $L = 3$. The ferromagnetic state therefore needs the full depth and carries the most noise channels, which is a second reason it suffers most.

5 Error mitigation

5.1 Zero-noise extrapolation. We ran Richardson extrapolation on every correlator C_{ij} and every $\langle X_i \rangle$ evaluated at λp

for $\lambda = 1, 2, 3$ (linear noise scaling), holding out the $p = 0$ truth, and rebuilt $S(q)$ before classifying. At $p = 0.01$ this is effective (Figure 2): the median $|\Delta C_{ij}|$ falls from 0.063 to 0.011, $|\Delta m_x|$ falls from 0.26 to 0.023, and label agreement with the clean circuit rises from 70% to 98%. The threshold boundaries return to within $|\delta h| \leq 0.09$ with a median of 0.03; at $\kappa = 0.93$, for example, the shift goes from -0.48 to -0.04. The price is a variance-amplification factor of $\sum \gamma^2 = 19$. At $p = 0.05$ zero-noise extrapolation fails for both linear and composed-channel scaling, because the remaining signal is about 1% and the extrapolation has to reach out to $p = 0.15$.

5.2 Parity verification. Post-selecting on the parity operator $P = \prod_i X_i$ accepts 71% of the weight at $p = 0.01$ and 51% at $p = 0.05$. It lowers the error in C_{ij} only from 0.063 to 0.053, because most errors stay inside the even-parity sector.

5.3 Noise-aware re-optimisation. As a separate protocol on the cuts $\kappa = 0.2, 0.3, 0.7$ and 0.8 , with 8 values of h each, we minimised $\text{Tr}(H\rho_p)$ with exact gradients through the channels. This lowers the noisy energy by 0.2 to 1.0 per cut, but the order parameters move by only -0.016 to +0.028 and their error against exact diagonalisation is unchanged; at $\kappa = 0.8$ and $p = 0.01$, for example, it goes from 0.225 to 0.198. The optimiser gains energy mainly through m_x , which rises by +0.03 to +0.09, by leaving the ground state: at $\kappa = 0.2, h = 0$ the clean energy of the re-optimised parameters is -5.83, against -6.40 for the true ground state. Energy-optimal noisy states are therefore not diagnostic-optimal.

6 Conclusions and limitations

The clean phase diagram is reproduced accurately on rings of 8 to 12 sites. For a symmetric circuit accurate enough to represent these ground states, which costs 128 CNOTs, noise at $p = 0.01$ already removes the ferromagnet under fixed thresholds while the boundary positions defined by derivatives stay put, and $p = 0.05$ leaves no usable signal at all. Zero-noise extrapolation recovers the $p = 0.01$ map almost fully. The limitations are the small system sizes, the approximate overlay curves, the masking of the $\kappa \approx 0.5$ region for preparation quality, the use of the fixed-parameter protocol on the full grid with re-optimisation only on cuts, and the absence of shot noise in these exact simulations.

References. Full entries are in the repository’s literature notes. [S01] Organiser handout; [S07] starter-kit audit; [S10] Beccaria, Campostrini, Feo, PRB 73, 052402 (2006); [S12] Cea et al., arXiv:2402.11022 (2024); [S13] Beccaria, Campostrini, Feo, PRB 76, 094410 (2007); [S19] Temme, Bravyi, Gambetta, PRL 119, 180509 (2017); [S21] Bonet-Monroig et al., PRA 98, 062339 (2018); [S23] Sandvik, AIP Conf. Proc. 1297, 135 (2010); [S28] our channel-response derivation; [S30] Wecker, Hastings, Troyer, arXiv:1507.08969 (2015). Tools: PennyLane 0.44.1, NumPy, SciPy, Matplotlib.